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GENERIC UNIVERSITY
GENERIC College of Engineering and Technology
Department of Information Technology

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Subject: Software Engineering

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

for

Smart Canteen Ordering and Token System (SCOTS)

Version 2.3 (Approved)

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1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed Software Requirements Specification (SRS) for the "Smart Canteen Ordering and Token System (SCOTS)". This document describes the functional and non-functional requirements of the proposed system and is intended to serve as a reference for the development team, project guide, and evaluators. It is written for the stakeholders who will be involved in using, testing, or approving the system. Its structure is based on IEEE Std 830-1998 [1] and the practices described in [2].

1.2 Scope

This system allows students from any branch or year, as well as all college teaching and non-teaching staff, to browse the canteen menu, pre-order food, schedule a pickup Time Window later in the same operating day, and make digital payments. With the help of this system, Customers receive a token number for quick pickup, while Canteen Staff can manage orders via a dashboard, and the Canteen Admin can track daily sales. The system supports both online pre-ordering via the Customer mobile app and walk-in offline ordering directly at the counter managed by Canteen Staff. The system's scope is strictly limited to a single campus outlet and does not include delivery features.

1.3 Overview

This project presents the development of a mobile-based food ordering platform with a web dashboard for canteen staff and admin designed to streamline canteen operations and enhance the dining experience for college students and staff.

The system introduces features such as digital menu browsing, remote pre-ordering, secure online payments, and real-time order tracking through a dedicated token generation system.

The primary objective is to optimise the brief recess hours by minimising manual checkout delays, while simultaneously empowering canteen management with a dashboard for order handling and daily sales reporting.

1.4 Definitions, Acronyms, and Abbreviations

A. Document and System

- **SCOTS**: Smart Canteen Ordering and Token System, the proposed digital canteen ordering and queue management platform.
- **User:** Any person using the system, including a Customer, Canteen Staff member, or Canteen Admin (these three roles are defined in Section 2.1).
- **FR**: Functional Requirement, a statement of a specific behaviour or function the system must perform.

- **ETA**: Estimated Time of Arrival; in SCOTS, the predicted clock time at which an order will be Ready.

B. Orders

- **Immediate Order:** An order placed through the SCOTS app for the soonest possible pickup, as opposed to a Scheduled Order.
- **Scheduled Order:** An order placed for a future pickup Time Window (defined later in this section) later in the same operating day, instead of the soonest-possible pickup. A Scheduled Order is always prepaid by UPI.
- **Offline Order (Counter Order):** An order placed in person at the canteen counter where Canteen Staff enters item choices on behalf of a walk-in Customer, records Cash or UPI payment, shows the simulated ETA before payment, and issues a printed token receipt. An Offline Order does not require a Customer app account. Offline Orders enter the kitchen simulation immediately as Immediate Orders.
- **Token:** A number unique within an operating day, assigned to an order and used to identify it at pickup (see FR9).

C. Order Status

- **Received**: The initial order status assigned immediately after a Customer places an order and payment is confirmed (or Cash-at-Pickup is chosen). For Immediate Orders, it enters the active FIFO kitchen queue immediately; for Scheduled Orders, it remains in the system's hidden queue until its Promised Release Time.
- **Preparing**: The status of an order when kitchen preparation of the ordered items begins. Transitioning to this state triggers late-cancellation penalty rules for cancellations (per FR20.1/FR20.2).
- **Ready:** The status updated by Canteen Staff when the food is fully prepared and packaged. Marking an order "Ready" updates the digital display board, pushes a pickup notification to the Customer, disables self-cancellation (per FR20), and, if applicable, starts the auto-expiry countdown (default 20 minutes, per FR24/FR21/FR21.1).
- **Collected:** The final terminal status marked by Canteen Staff when the Customer presents their token (the app token or digital receipt for app orders, or the printed token receipt for Offline Orders) at the counter, receives their food and, for Cash-at-Pickup, pays the outstanding amount in full.
- **Cancelled:** The status of an order cancelled by the Customer, by Canteen Staff at the counter, or by the system under FR5.1 or FR7. Customer and Canteen Staff cancellations are subject to FR20 and its stated exceptions.
- **Expired:** The status of a Ready order that was not collected within the auto expiry window, applying the policy in FR21 and FR21.1.

D. Payment and Policy

- **Cash-at-Pickup:** A cash-on-pickup payment option where a Customer places an order via the app and pays physically in cash at the canteen counter upon order fulfilment. Available for Immediate Orders only.
- **UPI**: Unified Payments Interface, a real-time payment system used in India.
- **Razorpay:** The third-party payment gateway used for UPI payments and refunds; the system verifies payment status through outbound API requests, with an optional webhook backup (see Section 5.4).
- **Pending Due:** An outstanding balance added to a Customer's account when a Cash-at-Pickup order expires or is cancelled with a penalty, which must be cleared before the Customer can place further orders.
- **Auto-expiry window:** The time after an order is Ready (or, for a Scheduled Order, after the later of its Time Window start and its Ready time) within which the Customer must collect it. Default 20 minutes, configurable by the Canteen Admin (FR24, FR21).
- **No-Show**: An instance where a Customer fails to collect an order marked "Ready" within the auto-expiry window (default 20 minutes, per FR24) of its readiness or, for a Scheduled Order, within that window of the later of its Time Window start time and its "Ready" time.
- **Payment Pending / Payment Under Review:** Payment states, not order statuses, shown while a payment is being reconciled (Section 4.2). An order is not Received until its payment is confirmed.

E. Kitchen and Scheduling

- **Operating Day:** A single calendar day on which the canteen is open (9:00 AM to 6:00 PM, all days except Sundays and official college holidays).
- **Kitchen Queue (Active Queue):** The ordered list of accepted orders released for preparation. It contains Immediate/Offline Orders and released Scheduled Orders, arranged according to FR12.5 while preserving FIFO order among Immediate/Offline Orders. Unreleased Scheduled Orders remain outside the visible active queue.
- **FIFO:** First In, First Out. Immediate and Offline Orders retain their relative acceptance order. Scheduled Orders join the kitchen processing sequence at positions determined by FR12.5; their original placement timestamp does not by itself determine their position among active orders.
- **Checkout ETA:** The predicted clock time at which an Immediate Order would be Ready, calculated from the proposed cart at checkout using the scheduling simulation in Appendix B.3 and shown to the Customer for confirmation before payment or order

acceptance. It is not a guarantee; the Token ETA, defined below, is the locked commitment.

- **Token ETA (Locked ETA):** The fixed clock time at which an Immediate Order is predicted to be Ready, calculated at acceptance using the same scheduling simulation as the Checkout ETA. It does not change after acceptance.
- **Time Window:** The pickup interval selected for a Scheduled Order (default length 15 minutes).
- **Promised Release Time:** The time at which a Scheduled Order is due to enter the Kitchen Queue, calculated as **the start of its selected Time Window minus its preparation time**. It is fixed when the order is placed and is never recalculated.
- **Max Slack:** The maximum time, in minutes, by which a Scheduled Order's preparation start may be delayed past its Promised Release Time to accommodate Immediate or Offline Orders. Configurable by the Canteen Admin (default 15 minutes).
- **Latest Start Time:** The latest moment a Scheduled Order may begin preparation: **Promised Release Time plus Max Slack**.
- **Delay Grace:** The delay, from the Promised Release Time, up to which a Scheduled Order's Customer receives only a revised-ETA notification (default 15 minutes, configurable, not more than Max Slack).
- **Delayed:** A Scheduled Order whose actual kitchen start occurs after its Promised Release Time. The delay equals **actual start time minus Promised Release Time**. Delayed is a notification state, not a lifecycle status.

F. Interfaces

- **GUI**: Graphical User Interface, the visual interface through which users interact with the system.
- **Staff Dashboard / Admin Dashboard:** The web-based interfaces used by Canteen Staff and Canteen Admin, respectively, to manage orders and system settings.

G. Technical and Compliance

- **API**: Application Programming Interface, a set of rules that allows software applications to communicate with each other.
- **HTTPS**: Hypertext Transfer Protocol Secure, an encrypted protocol for secure communication over a network.
- **TLS**: Transport Layer Security, a cryptographic protocol that provides secure communication over a network.
- **3NF**: Third Normal Form, a level of database normalisation.

- **DFD**: Data Flow Diagram, a graphical representation of how data moves through a system.
- **ERD**: Entity-Relationship Diagram, a visual representation of database entities and their relationships.
- **DPDP Act:** Digital Personal Data Protection Act, 2023 (India), the law governing the handling of personal data.
- **HMAC**: Hash-based Message Authentication Code, a cryptographic check used to verify payment signatures.
- **FCM**: Firebase Cloud Messaging, the push-notification service used for Android and iOS.
- **APNs**: Apple Push Notification service, used by FCM to deliver notifications to iOS devices.
- **SMTP**: Simple Mail Transfer Protocol, used to send verification and password-reset emails.
- **WebSocket**: A persistent two-way network connection used for real-time updates.

2. General Description

2.1 User Characteristics

- **Customers**: Students or teaching and non-teaching staff of the college with basic smartphone literacy; they will use the app to browse the menu and place orders.
- **Canteen Staff**: Responsible for accepting, preparing, and marking orders as ready; assumed to have minimal technical background, so the interface must be simple.
- **Canteen Admin**: The canteen owner or designated manager; manages the menu, prices, Canteen Staff accounts, and policy settings, and views daily sales reports.

2.2 User Workflows

This section explains how to use SCOTS from the perspective of each type of user, so that the interface and workflow described in Sections 3 and 5 are grounded in a concrete step-by-step usage flow.

2.2.1 For Customers (Placing an Order)

Steps	Process
1.	**Register/Login:** First-time Customers enter their name, Enrolment number/College ID, college email address and phone number, create a password, and verify their email address. Registered Customers log in using their verified college email address and password.
2.	**Browse the Menu:** On the Home screen, view the day's available menu items along with prices and static estimated preparation time. Items marked "Sold Out" cannot be ordered.
3.	**Add Items to Cart:** Add items and change quantities. The app shows each item's static estimated preparation time. The predicted Ready time for an Immediate Order is shown only at checkout (step 5).
4.	**Choose Immediate or Scheduled Pickup:** Either place the order for the soonest possible pickup or select a future Time Window later in the same operating day so the food is freshly prepared closer to when it can be collected. For an Immediate Order, the predicted Ready time is shown at checkout; you can return here and choose a Scheduled pickup instead.
5.	**Checkout:** Review the order and total amount. For an Immediate Order, the app shows the predicted Ready time (Checkout ETA); confirm it to continue, or go back and choose a Scheduled pickup. Then pay via UPI, or choose Cash-at-Pickup (Immediate Orders only; Scheduled Orders are UPI-prepaid only). Accounts restricted for repeated No-Shows (FR23.1) will only see the UPI prepayment option. Accounts suspended for repeated No-Shows (FR23.2) cannot proceed to checkout at all. Accounts with an outstanding Pending Due (FR22.1) cannot place new orders until it is cleared.

Steps	Process
6.	**Receive Your Token:** Once confirmed, a unique token number and a locked-in Token ETA (or scheduled pickup time) are generated and shown in the app. The receipt states the auto-expiry window; the exact expiry time is shown once the order is Ready.
7.	**Track Order Status:** Check the app for the current status (Received, Preparing or Ready). A Customer may cancel before Ready, subject to the applicable refund or penalty rules in FR20. A configured number of counted Received-stage self-cancellations in one operating day (default two) blocks new orders for 24 hours from the threshold-reaching cancellation (FR20.4).
8.	**Collect Your Order:** When the order is Ready, show the token or digital/printed receipt in the app at the counter. If Cash-at-Pickup (for Immediate Orders) was selected, pay the outstanding amount before receiving the food.
9.	**Avoid No-Shows:** If an order marked "Ready" is not collected within the auto-expiry window (default 20 minutes), it expires, the amount is handled under FR22, and a No-Show is logged. Repeated No-Shows restrict, then suspend, the account (FR23).

2.2.2 For Canteen Staff

Steps	Process
1.	**Login:** Log in to the Staff Dashboard using staff credentials.
2.	**Take Walk-in Order:** Take the item selections of a walk-in Customer, enter them on the Staff Dashboard, choose the payment method (Cash or UPI), collect payment, and issue the printed token receipt (FR10, FR11.2).
3.	**View Incoming Orders:** Canteen Staff view orders in their current active-queue processing order. Immediate and Offline Orders retain FIFO order. Scheduled Orders appear only after release at their Promised Release Time (FR15, Appendix B.3).
4.	**Update Order Status:** Update each order's status from Received → Preparing → Ready. Each status update asks for confirmation in a confirmation box ("Mark order #<token> as <new_status>?") to prevent accidental misclicks. Marking an order "Ready" updates the token display board and notifies the Customer automatically once staff confirm.
5.	**Handle Expired Tokens:** If a token expires due to non-collection, it is automatically removed from the active queue; staff should set aside or discard the corresponding food item as per canteen policy.
6.	**Mark Order as Collected:** Verify the token and full payment, record any Cash-at-Pickup payment under FR13.1, hand over the food, and mark the order Collected.

2.2.3 For Canteen Admin

Steps	Process
1.	**Login:** Log in to the Admin Dashboard using admin credentials.
2.	**Manage Menu:** Add, edit, or remove menu items, set each item's approximate preparation time, and update prices. Mark items "Sold Out" if ingredients run out.
3.	**Configure Scheduled-Pickup Time Windows:** Set the kitchen cook count N, Time Window length, Scheduled Order capacity share, Max Slack and Delay Grace under FR24. The default Scheduled Order share is two-thirds of physical cook capacity; Immediate/Offline Orders use the remaining share.
4.	**View Sales Reports:** Generate a daily sales report showing total orders, revenue, and most-ordered items.
5.	**Review No-Show Log:** Check which Customer accounts have been auto-restricted or suspended for repeated No-Shows and manually lift restrictions or reinstate accounts after reviewing a case.
6.	**Manage Staff Accounts:** Create, update, and deactivate Canteen Staff login accounts (FR2.2).
7.	**Configure Policies:** Set the policy values listed in FR24 (e.g., No-Show forfeiture, late-cancellation penalty, auto-expiry window, minimum lead time, ETA reconfirmation tolerance).
8.	**View Audit Log:** Review the audit log of order, refund, Pending Due, and account-restriction actions (Section 4.1).

3. Functional Requirements

3.1 Description

The major functional requirements of SCOTS are listed below. Requirements are grouped by feature area (3.1.1 to 3.1.11).

3.1.1 User Accounts & Access

- **FR1:** The system shall allow a Customer to self-register by providing their name, college email ID, Enrolment number (students) or College ID (college teaching and non-teaching staff), phone number and password. The account shall become active only after verification of the registered college email address. The Customer shall log in using that email address and password. The email ID and the Enrolment number/College ID must each be unique across accounts, to prevent duplicate account creation. Canteen Staff accounts are created and managed exclusively by the Canteen Admin (FR2.2) and are not self-registerable.
- **FR2:** Canteen Staff and Canteen Admin accounts shall be provisioned and managed as specified in FR2.1 and FR2.2.
- **FR2.1:** Authorised college IT personnel shall provision and deactivate Canteen Admin accounts outside the Customer and Canteen Staff interfaces. At least one active Canteen Admin account shall exist at all times.
- **FR2.2:** A Canteen Admin shall be able to create, update, and deactivate login accounts for Canteen Staff. Newly created Canteen Staff accounts shall receive a single-use activation link at their registered email address to set their password. Deactivation shall invalidate active sessions without deleting historical orders or audit records.
- **FR3:** The system shall provide a password reset option for all users through a single-use, expiring reset link sent to the registered email address. The response shall not reveal whether an email address is registered, and reset and activation requests shall be rate-limited.

3.1.2 Menu Display & Management

- **FR4:** The system shall display the daily menu along with prices, food pictures, ingredients, and each item's static estimated preparation time (distinct from the Checkout ETA described in FR8).
- **FR4.1:** The system shall allow Customers to search the menu by item name and filter menu items by category.
- **FR5:** The system shall allow the Canteen Admin to add, edit or remove menu items and mark items "Sold Out". Sold Out items cannot be added to a cart. Removing an

item shall not alter existing orders or audit records. If an item can no longer be fulfilled, the Canteen Admin shall use FR5.1.

- **FR5.1:** If the Canteen Admin marks an item "Sold Out", the system shall cancel every Received order (Immediate, Offline or Scheduled) containing it, refund all amounts paid in full, void unpaid Cash-at-Pickup commitments, release unreleased Time Window capacity (FR20.3, Appendix B.4), remove released orders from the active queue and notify affected Customers (for an unlinked Offline Order, Canteen Staff shall process any cash refund). These cancellations do not count toward FR20.4.

3.1.3 Ordering & Checkout

- **FR6:** The system shall allow a Customer to add items to a cart, change quantities, remove items and submit the cart. Before confirmation, the server shall revalidate availability, prices, account eligibility and scheduling constraints; if the price or cart changes, the Customer shall confirm the revised total. Each confirmed order shall keep a snapshot of item names, quantities and prices.

- **FR6.1:** The system shall enforce a configurable maximum preparation time per Immediate or Offline Order (default 12 minutes, not above Max Slack). If a cart exceeds it, the Customer or Canteen Staff shall reduce the cart, split it (each part passing the FR12.5 check) or choose a Scheduled Order.

- **FR7:** The system shall accept new orders only during operating hours (9:00 AM to 6:00 PM IST on operating days). An order shall be accepted only if it can be Ready at least one auto-expiry window before closing (a Scheduled Order's Time Window must also satisfy FR12.1–FR12.4). If an accepted order can no longer meet this, the system shall cancel it, refund any amount paid in full and record no No-Show or Pending Due. No collection extension shall be offered.

- **FR8:** Before an Immediate Order is paid for or accepted, the system shall simulate the cart (Appendix B.3) and show the predicted Ready clock time (Checkout ETA) on the checkout screen. The Customer shall confirm it before payment starts (UPI) or the order is accepted (Cash-at-Pickup), or may return to the cart or choose a Scheduled pickup. When the Customer selects Pay or Place Order, the system shall re-run the simulation; if the Ready time differs by more than the configured reconfirmation tolerance (default 2 minutes), the Customer shall reconfirm the revised time. The Token ETA shall be locked upon acceptance.

- **FR9:** The system shall assign every confirmed order a globally unique order ID and a token number unique within its operating day. Daily token numbering shall restart at opening without reusing an order ID. App-based orders shall display the token in the Customer's app; Offline Orders shall provide it on the printed receipt.

- **FR10:** The system shall allow Canteen Staff to place an Offline Order through the Staff Dashboard by selecting items, showing the ETA, recording Cash or UPI payment, generating a token and printing a receipt (FR11.2). Offline Orders enter the kitchen simulation as Immediate Orders and are exempt from FR20.4, FR23.1 and FR23.2. Cancelling an Offline Order while Received refunds 100%; while Preparing, or on No-Show expiry, the configured retention percentage (default 100%) is kept and any remainder refunded.
- **FR11:** The system shall provide a receipt for every confirmed order, as a digital receipt (FR11.1) and a printed token receipt (FR11.2).
- **FR11.1:** Upon order confirmation, the system shall generate a digital receipt for the linked Customer showing canteen name, order ID, token number, date and time, items with quantity and price, total, payment method and status, Token ETA or Time Window, and the auto-expiry window. It shall state that the expiry countdown starts only when the order is marked Ready (for a Scheduled Order, at the later of the Time Window start and the Ready time), so the exact expiry time is shown once the order is Ready (FR18.1, FR19).
- **FR11.2:** The system shall print a physical token slip/receipt at the counter containing the required details in FR11.1. For Offline Orders, printing shall start through the browser print dialog after staff records payment confirmation. For app-based orders, Canteen Staff may print a receipt on request during collection.

3.1.4 Scheduled Ordering System (Capacity Slotting / Time-Block Reservation Algorithm)

- **FR12:** Scheduled ordering (prepaid by UPI only) shall be supported as specified in FR12.1 to FR12.7.
- **FR12.1 (Minimum Lead Time):** The system shall offer a Time Window only if it starts at least the configured minimum lead time (default 30 minutes) from now and the proposed order's predicted completion (Appendix B.3) does not exceed the window start. Other windows shall be disabled and the nearest eligible later window suggested.
- **FR12.2 (Time Window Selection):** The system shall allow a Customer, at the time of ordering, to optionally select a predefined pickup Time Window (e.g., 15-minute intervals) later in the same operating day, instead of the soonest-possible pickup.
- **FR12.3 (Time Window Capacity Limits):** Each Time Window shall have a Scheduled Order capacity in preparation-work minutes: **the number of cooks (N) × the Time Window length × the configured Scheduled Order share** (default 2/3, strictly between 0 and 1). The remainder is reserved for Immediate/Offline Orders (Appendix B.1).

- **FR12.4 (Availability Validation):** The system shall atomically validate and reserve Time Window capacity before confirming an order. Before UPI payment it shall hold a temporary reservation (default 3 minutes), released on failure or expiry. Verified payment converts a valid reservation into a confirmed order exactly once; a late payment shall trigger a fresh eligibility check, with a full refund if the order cannot be accepted. If a window lacks capacity, the nearest alternative shall be suggested.
- **FR12.5 (Dynamic Queue Integration and Max Slack):** At placement, the system shall fix **Promised Release Time = Time Window start – preparation time**, and **Latest Start Time = Promised Release Time + Max Slack** (default 15 minutes, Admin-configurable, not above the Time Window length). The scheduler shall assign each order to the cook who becomes free earliest (Appendix B.3). If placing an Immediate/Offline Order ahead of a Scheduled Order would push it past its Latest Start Time, the Immediate/Offline Order shall be placed after it and the schedule recomputed for every later affected Scheduled Order. Immediate/Offline Orders are never rejected by this check; their Token ETAs include the resulting position. The check applies independently to each part of a split cart.
- **FR12.6 (Single ETA Simulation):** The Checkout ETA and Token ETA shall both use the single simulation defined in FR8 and Appendix B.3. No separate calculation shall be used for either value.
- **FR12.7 (Delay Handling):** For a Scheduled Order, **delay = actual kitchen start – Promised Release Time**. When the order starts, the system shall notify the Customer of its revised Ready ETA. If the delay exceeds the Delay Grace snapshotted at placement (default 15 minutes, not above Max Slack), the system shall also offer cancellation for a 100% refund, available until the order is Ready and taking precedence over FR20 penalties.

3.1.5 Payments & Refunds

- **FR13:** The system shall support online payment through UPI as well as a "Cash-at-Pickup" cash option (for Immediate Orders only; Scheduled Orders shall be prepaid by UPI).
- **FR13.1:** For a Cash-at-Pickup order, Canteen Staff shall record the amount received and confirm full payment before handing over the food; the recording shall update the receipt and log the staff account and time. Canteen Staff shall keep a daily counter cash book of cash received and refunded. Collected shall be allowed only when the order is Ready and fully paid.
- **FR13.2:** Before confirming a UPI order or online Pending Due settlement, the server shall verify the Razorpay payment signature (HMAC) and fetch the payment status through the Razorpay API, checking amount, currency, order reference and captured

state. Client-reported success alone is not confirmation, and repeated or out-of-order callbacks shall not create duplicate orders, tokens or payments.

- **FR14:** The system shall track and issue refunds as specified in FR14.1 and FR14.2.
- **FR14.1:** The system shall track each refund as Pending, Succeeded or Failed with amount and payment reference. It shall not report a refund complete before gateway confirmation or recorded cash repayment, shall retry recoverable failures without duplication and shall flag unresolved failures for Canteen Admin review.
- **FR14.2:** UPI refunds shall be made against the original gateway transaction. Cash refunds shall be paid at the counter by authorised Canteen Staff and recorded with order ID, amount, staff account and time. Total refunds shall never exceed the refundable amount.

3.1.6 Kitchen Queue & Order Fulfilment

- **FR15:** The system shall notify the Staff Dashboard in real time when an order enters the active kitchen queue: on placement for Immediate Orders, at the Promised Release Time for Scheduled Orders (Appendix B.3). Held Scheduled Orders shall not appear until released.
- **FR16:** The system shall allow Canteen Staff to update order status from Received to Preparing, from Preparing to Ready, and from Ready to Collected. Each update shall require confirmation in a box naming the order token and new status; dismissing it leaves the order unchanged. On confirmation, the system shall atomically validate the current order version and apply the update once; if a cancellation or expiry committed first, the update shall be rejected and the current status shown.

3.1.7 Order Tracking, Display & Notifications

- **FR17:** The system shall display, on a digital screen/board near the counter, all tokens in Ready status awaiting collection and the next 5 token numbers in active-queue order.
- **FR18:** The system shall allow a Customer to follow current orders and review past orders as specified in FR18.1 and FR18.2.
- **FR18.1:** The system shall allow a Customer to view the live status, token, Ready ETA and, once Ready, the exact auto-expiry time of each order linked to that Customer's account, subject to server-side ownership checks. An Offline Order appears in app history only if explicitly linked to the Customer.
- **FR18.2:** The system shall maintain and display a comprehensive order history for each Customer within the app, allowing them to view details of all past orders, including items purchased, total amount, payment method, order date, and final status (Collected, Cancelled or Expired).

- **FR19:** The system shall notify a linked Customer when an order is Received, Preparing, Ready, Collected, Cancelled, Expired or Delayed. The Ready notification shall state the exact auto-expiry time; the Delayed notification shall state that a Scheduled Order's expiry moves with its revised Ready ETA. Notifications shall be sent only after the status change is committed. For an unlinked Offline Order, the information shall be available to Canteen Staff and on the counter receipt.

3.1.8 Order Cancellation

- **FR20:** The system shall allow a Customer to self-cancel any order before it reaches Ready, subject to FR20.1 and FR20.2; ordinary cancellation is disabled after Ready. System-initiated cancellations (FR5.1, FR7) and eligible delay cancellation (FR12.7) take precedence over penalties. Canteen Staff may cancel an Offline Order before Ready at a walk-in Customer's request under FR10.

- **FR20.1:** For prepaid orders: if "Received", refund 100% with no penalty; if "Preparing", apply the late-cancellation penalty (default 100% of the order total, configurable) and refund any remainder. Offline Orders follow FR10.

- **FR20.2:** For Cash-at-Pickup orders: if cancelled while "Received", void with no penalty; if "Preparing", add the late-cancellation penalty (configured percentage, default 100%) to the Customer's Pending Due (FR22.1).

- **FR20.3:** Upon cancellation, an unreleased Scheduled Order shall return its preparation time to its Time Window's capacity (FR12.3). A released Scheduled Order (status "Received" in the active queue, or "Preparing") and any Immediate/Offline Order shall be removed from the active queue without returning Time Window capacity, and the cook-allocation simulation shall be recalculated immediately (Appendix B.3).

- **FR20.4:** If a Customer self-cancels the configured number of orders (default two) at the Received stage in one operating day, the system shall disable ordering for 24 hours from the threshold-reaching cancellation. Cancellations under FR5.1 and FR12.7 do not count. The block does not clear a Pending Due or suspension, and accepted orders remain valid.

- **FR20.5:** Customer cancellation, staff status update and automatic expiry for the same order shall be serialised atomically against the latest committed status. A cancellation after Preparing may proceed with the applicable penalty; after Ready it shall be rejected unless FR5.1 or FR7 applies. An order already Cancelled or Expired shall never be restored.

3.1.9 Collection, Expiry & No-Show Policy

- **FR21:** If an order marked Ready is not collected within the configured auto-expiry window (default 20 minutes), the system shall mark it Expired and free the queue slot.

The countdown starts when the Ready status is committed. Release and expiry deadlines shall be persisted and rebuilt after a server restart without duplicating releases, penalties or notifications.

- **FR21.1 (Scheduled Order Expiry):** For a Scheduled Order, the expiry countdown starts at the later of the Time Window start and the Ready time. FR21 to FR23 otherwise apply, and, being prepaid, a Scheduled Order never creates a Pending Due.
- **FR22:** For a prepaid or fully paid order that expires due to non-collection, the system shall forfeit the configured percentage of the amount paid (default 100%) and refund any remainder. For an unpaid Cash-at-Pickup order that expires, the system shall add the configured percentage of the unpaid order amount (default 100%) to the Customer's Pending Due. A fully paid order shall not create a Pending Due.
- **FR22.1:** The system shall not permit a Customer with an outstanding Pending Due to place new orders until it is cleared online or at the counter. Each due keeps its creation time and amount; if unpaid at the review deadline (default 3 calendar days, Asia/Kolkata), the Canteen Admin shall be alerted and the account flagged for review, without automatic suspension. Settlement shall reduce the balance exactly once and clear only the Pending Due block.
- **FR23:** The system shall escalate repeated No-Shows as specified in FR23.1 to FR23.3.
- **FR23.1:** For each Customer (excluding Offline Orders), the system shall count No-Show events within the configured lookback period (default 30 days). At two, the Customer shall be restricted to UPI prepayment; the restriction lifts automatically when the count falls below two or events are waived under FR23.3.
- **FR23.2:** Using the configured lookback period defined in FR23.1, if a Customer's No-Show count reaches three while the Customer is restricted under FR23.1, the system shall suspend ordering privileges. The suspension shall remain until the Canteen Admin reviews and reinstates the account.
- **FR23.3:** The system shall provide the Canteen Admin with a No-Show log of restricted and suspended accounts. After review, the Admin may lift a restriction or reinstate an account and shall record a reason; the responsible No-Show events are then waived for future counts but kept in the audit history. This does not clear Pending Dues or unrelated blocks.

3.1.10 Administration & Reporting

- **FR24:** The Canteen Admin shall be able to configure the policy settings listed in Appendix B.5. The system shall validate **$0 < \text{Scheduled Order capacity share} < 1$**, **$\text{Delay Grace} \le \text{Max Slack} \le \text{Time Window length}$**, **$\text{preparation-time limit} \le \text{Max Slack}$** and **$1 \le N \le 5$**. Changes shall apply only to new orders (cook-count changes

from the next operating day), accepted orders shall keep snapshots of their applicable values, and all changes shall be audit-logged.

- **FR25:** The system shall generate a daily sales report showing order counts by status, gross payments, refunds, net receipts, retained cancellation/No-Show amounts and outstanding Pending Dues, split by cash/UPI, channel (app/counter) and order type (Immediate/Scheduled), plus most-ordered items. Collections and refunds shall be attributed to the date they are recorded. Unpaid orders and Pending Dues shall not count as money received.
- **FR26:** The system shall allow the Canteen Admin to view and filter the audit log by date, actor, order/account reference and action type. Entries shall show timestamp, outcome and before/after values without credentials, and shall not be editable.

3.1.11 Help & Support

- **FR27:** The system shall provide dedicated Help, FAQ, and Contact sections detailing system rules, resolving payment failures, explaining how a Customer can request Canteen Admin review of a restriction or suspension (FR23.3), and communicating all other administrative policies.

3.2 Technical Issues

A few technical challenges are anticipated during the development and operation of the system, and are noted here so that the design can account for them:

- Handling multiple orders placed at the same time during the recess rush, without token number clashes or database locking issues.
- Keeping the token display screen synchronised with the server in real time.
- Dependence on the college Wi-Fi network for online Customer orders; the Customer app shall show a connection-lost state and retry automatically, but shall not place an order or payment without server confirmation (FR6, FR12.4). During brief Wi-Fi interruptions, Canteen Staff can still accept walk-in Cash Offline Orders over the wired campus LAN (Section 5.2); UPI payments and real-time updates need an active connection (FR13.2).
- Preventing a race condition where a Customer cancellation request (FR20) and a Canteen Staff status update to "Preparing" for the same order are submitted at nearly the same moment; this shall be resolved atomically according to FR20.5.
- Managing parallel FIFO and Scheduled Order streams by dynamically interleaving preparation times. To prevent server overload, this requires an event-driven and timer-driven release logic (FR12.5) for scheduled releases rather than continuous time-based polling.

4. Non-Functional Requirements

4.1 Security

- Passwords shall be stored using a one-way hashing algorithm (e.g., bcrypt); plain-text passwords shall never be stored.
- Role-based access control shall ensure that only authenticated Canteen Staff and Canteen Admin accounts can use the dashboards and that a Customer can access only their own orders, receipts, payments and Pending Dues. Knowing an order ID or token shall not grant access, and deactivation or suspension shall be enforced by the server.
- Payment credentials shall not be stored on the college server; only the transaction metadata needed for reconciliation, receipts and audit may be kept, and gateway secrets shall never appear in client code or logs.
- All user actions that create, change or cancel orders, refunds, Pending Dues or account restrictions, including Canteen Staff and Canteen Admin actions, shall be recorded in an audit log that cannot be edited through the application and is viewable by the Canteen Admin.
- Login sessions shall expire after a configurable period of inactivity, and an account shall be temporarily locked after a configurable number of failed login attempts.
- Customer and staff personal data shall be collected only as needed for ordering, payment and account management, and shall be handled in line with the Digital Personal Data Protection Act, 2023 (India) [3].

4.2 Reliability

The system shall correctly process an eligible, valid order and generate its token in at least 99.9% of attempts under documented test conditions. If payment succeeds but confirmation is interrupted, the system shall reconcile and either confirm the order exactly once or fully refund it, showing Payment Pending or Payment Under Review meanwhile.

4.3 Availability

The system shall achieve at least 99% availability during scheduled canteen operating minutes in each calendar month. Availability means that an eligible Customer can access the menu and submit a valid order and Canteen Staff can retrieve and update accepted orders. Outages during operating hours count as unavailable time; scheduled closed days are excluded from the denominator. Authenticated Canteen Admin access shall also be available from 8:30 AM for pre-opening work, except during announced maintenance.

4.4 Maintainability

The system shall be developed using a modular architecture (separate modules for user management, ordering, payments, and reporting) so that individual modules can be updated or

debugged without affecting the rest of the system. Code shall be documented with comments, and a basic developer guide shall be maintained.

4.5 Reusability

The core modules of SCOTS (menu management, ordering, and token generation) should be designed generic enough to be reused for other similar use cases on campus, such as a library book-issue token system or a stationery/print-shop ordering system, with minimal modification.

4.6 Data Backup & Recovery

The system shall perform an automated daily backup of the production database (user, menu, order and transaction records) to a location separate from the primary server. In the event of data loss or corruption, the system shall support restoration from the most recent backup within 4 hours of a reported failure. The maximum tolerated data loss is 24 hours. Manual recovery shall proceed in five steps: (1) restore the latest backup and set the next token number above the highest token recorded for that day; (2) match gateway payments to database records; (3) add missing cash orders from the counter cash book; (4) flag orders with unknown status for Canteen Admin review, with no automatic charge or No-Show; and (5) resume ordering after reconciliation.

4.7 Usability

Given the minimal technical background assumed of Canteen Staff (Section 2.1), the Staff Dashboard shall be designed so that a staff member can submit an order status update (e.g., Received → Preparing → Ready) in no more than 2 taps/clicks and within 3 seconds, without consulting a manual. The 3-second target measures action submission, including the confirmation step.

5. Interface Requirements

5.1 GUI

The Customer mobile app for Android 8.0 and later and iOS 15.0 and later, and the web dashboards, shall provide clear, simple and responsive interfaces. Customer screens shall include Login/Register and Password Reset, Menu/Home with search and category filters, Cart and Checkout with Immediate or Scheduled pickup selection and Checkout ETA confirmation, Order Status/Token, Order History, Pending Due and settlement, payment/refund status, and Help/FAQ/Contact. Canteen Staff screens shall include Order Queue, Walk-in Order Entry, cash collection, due settlement, cash refund and Receipt Print controls. Canteen Admin screens shall include Menu Management, Staff Account Management, Policy Settings including Time Window and cook configuration, Daily Sales Report, No-Show Log, account review, Refund Review (FR14.1), Pending Due Review (FR22.1) and Audit Log.

5.2 Hardware Interface

- A token display unit (an LED/LCD screen or a low-cost monitor) mounted at the canteen counter and driven by the counter mini-PC, which also runs the Staff Dashboard. The mini-PC connects to the college server over the campus network (wired LAN).
- A receipt printer connected to the counter mini-PC. Receipts are printed from the Staff Dashboard through the browser print dialog, as specified in FR11.2.
- Customer smartphones used to access the mobile app, meeting the minimum and acceptance-test device profiles in Section 7.
- An on-campus college server to host the application and database.

5.3 Software Interface

- **Backend:** Node.js [4] with Express.js [5] and a WebSocket library (e.g., Socket.IO or ws), supporting the asynchronous event handling used for queue updates, ETA calculation and real-time connections.
- **Database:** PostgreSQL [6], supporting concurrent order transactions, time-based scheduling and the configured rolling No-Show lookback.
- **Frontend (Customer Mobile App):** Flutter [7] for the Customer mobile app, targeting Android 8.0 and later and iOS 15.0 and later.
- **Frontend (Staff Dashboard and Admin Dashboard):** React.js [8], chosen for a fast, responsive web-based management dashboard and point-of-sale interface for Canteen Staff and Canteen Admin.
- **Payment Gateway:** Razorpay API [9], as specified in FR13.2 and Section 5.4. The integration shall provide a Sandbox environment for pre-production testing.

- **Push Notifications:** Firebase Cloud Messaging (FCM) for Android and iOS order-status notifications under FR19. iOS delivery shall use FCM with Apple Push Notification service (APNs).
- **Email Service:** An SMTP-based email service for account verification (FR1) and password-reset emails (FR3).

5.4 Communication Interface

- Client-Server: Communication between client applications and the backend server shall use HTTPS (TLS 1.2 or higher). For each UPI payment, the server shall verify the Razorpay HMAC signature and fetch payment status through an outbound Razorpay API request. The system shall not depend on an inbound webhook; a webhook may be used as an optional backup notification.
- Real-time Updates: The token display board and Staff Dashboard shall use secure persistent WebSocket connections and fall back to short-interval polling. Reconnecting Customers, Staff and display boards shall fetch authoritative current state. Push notifications prompt the app to refresh and are not the sole record of status changes.

6. Performance Requirements

- The system shall support at least 150 concurrent users during a 30-minute peak-recess test.
- Test conditions shall record server configuration, database size, client-device profiles, network latency/bandwidth, request mix and request rate. Test the Customer app on the device profiles in Section 7. Measure SCOTS processing before UPI initiation and after verifiable payment status separately from gateway and Customer interaction time.
- The 95th-percentile SCOTS processing time for placing a valid order, excluding time spent in the external UPI payment flow, shall not exceed 3 seconds under the documented 150-concurrent-user test conditions.
- The token display screen shall refresh within 2 seconds of an order status change being committed after staff confirmation (FR16).
- The system shall be able to process at least 500 orders per day at the current canteen scale.

7. Design Constraints and Assumptions

- The SCOTS application shall use open-source technologies where practical to keep the project within a student-project budget. Third-party services and Apple development/distribution prerequisites are listed below. Cost basis: Razorpay per-transaction gateway fees under its published merchant pricing, Firebase Cloud Messaging (free), an SMTP email service on a free or college-provided tier, the Apple Developer Program fee (US$99 per year) is the only mandatory recurring iOS cost. Enrolment is required before iOS acceptance testing.
- The application must run on existing college Wi-Fi infrastructure; no new networking hardware will be installed.
- The Customer app shall support Android 8.0+ and iOS 15.0+. Minimum acceptance-test profiles: Android 8.0 with 3 GB RAM at 720 × 1280, and iOS 15.0 on an iPhone 8-class device, plus one small-screen and one large-screen device per platform (e.g., Redmi 9A and Galaxy A14; iPhone SE (2nd generation) and iPhone 13 Pro Max).
- The database design must comply with normalisation principles (up to 3NF) to avoid redundancy.
- Assumes every Customer holds a valid college email ID and Enrolment number/College ID, and that every Canteen Staff and Canteen Admin has an email address able to receive activation and reset links.
- Assumes a Razorpay merchant account with Sandbox credentials is available before the integration and payment testing week in Section 10.
- Assumes the server (Section 5.2) is available outside canteen hours so the daily backup and recovery procedure in Section 4.6 can run.
- Kitchen capacity is modelled with N cooks (integer 1–5, default 2). N remains constant during each operating day; a cook-count change takes effect the next operating day. Cook absence during the day is not modelled. Actual kitchen overruns are handled under FR12.7.
- iOS release assumes access to an Apple Developer Program account, an APNs key, a Mac for iOS builds, and TestFlight or App Store distribution. The Apple Developer Program fee (US$99 per year) is the only mandatory recurring iOS cost. Enrolment in the Apple Developer Program is required before iOS acceptance testing.

8. Operational Scenarios

Scenario 1 - Immediate Ordering with Checkout ETA: A Customer browsing the app during a busy recess adds items to the cart and proceeds to checkout. The app shows the predicted Ready time (e.g., “3:42 PM”). Accepting this, the Customer confirms it, pays via UPI, and receives a token with a locked-in ETA. Instead of waiting in a physical line, the Customer tracks the status remotely and arrives at the counter when their token appears as Ready on the display board.

Scenario 2 - Kitchen Order Handling: Canteen Staff view incoming orders in active-queue processing order. The scheduler assigns each order to the cook who becomes free earliest and recalculates the schedule when an order becomes Ready or is cancelled. Staff mark an order Ready after preparation; the system updates the counter display and notifies the Customer once the status change is confirmed.

Scenario 3 - Menu Update: Before opening, the Canteen Admin uses the dashboard to update prices and mark out-of-stock items as "sold out," ensuring Customers only see accurate menu options before placing orders.

Scenario 4 - End-of-Day Reporting: At closing time, the Canteen Admin generates a daily sales report to reconcile cash and digital payments and analyse popular items for future inventory planning.

Scenario 5 - Order Placed but Not Collected (No-Show): If a Customer does not collect a Ready order within the auto-expiry window, the system expires it, applies the forfeiture or Pending Due rule (FR22), logs the No-Show, and applies restriction or suspension if the thresholds in FR23 are reached.

Scenario 6 - Scheduled Pickup: A Customer selects a Time Window and pays via UPI. The system fixes the Promised Release Time at placement. If Immediate or Offline Orders would push the order past its Latest Start Time, the scheduler places them after it (FR12.5). If the delay exceeds Delay Grace, the Customer is notified and may cancel for a full refund (FR12.7).

Scenario 7 - Voluntary Order Cancellation: A Customer cancels an order before Ready. The system processes the request atomically under FR20.5, applies the applicable refund or penalty, returns capacity only for an unreleased Scheduled Order, and records no No-Show.

Scenario 8 - Walk-in Offline Ordering: A walk-in Customer orders at the counter. Canteen Staff enter the order on the Staff Dashboard, show its ETA, record Cash or UPI payment and hand over the printed token receipt (FR10). The order is then handled as an Immediate Order.

9. System Design Diagrams

This section presents the diagrams applicable to the current scope of SCOTS: an Entity Relationship (ER) Diagram for the database design, a Use Case Diagram for the functional/actor view, Data Flow Diagrams (Context, Level 1, and Level 2) for the process view, and a Class Diagram for the object view.

9.1 Entity-Relationship (ER) Diagram

The ERD links Customer, Orders, Menu Item, Payment and Token, with Employee specialised into Staff and Admin. It shows customer ordering, staff updates and offline ordering, admin menu management, and order-item quantities and subtotals, alongside entity attributes and relationship cardinalities.

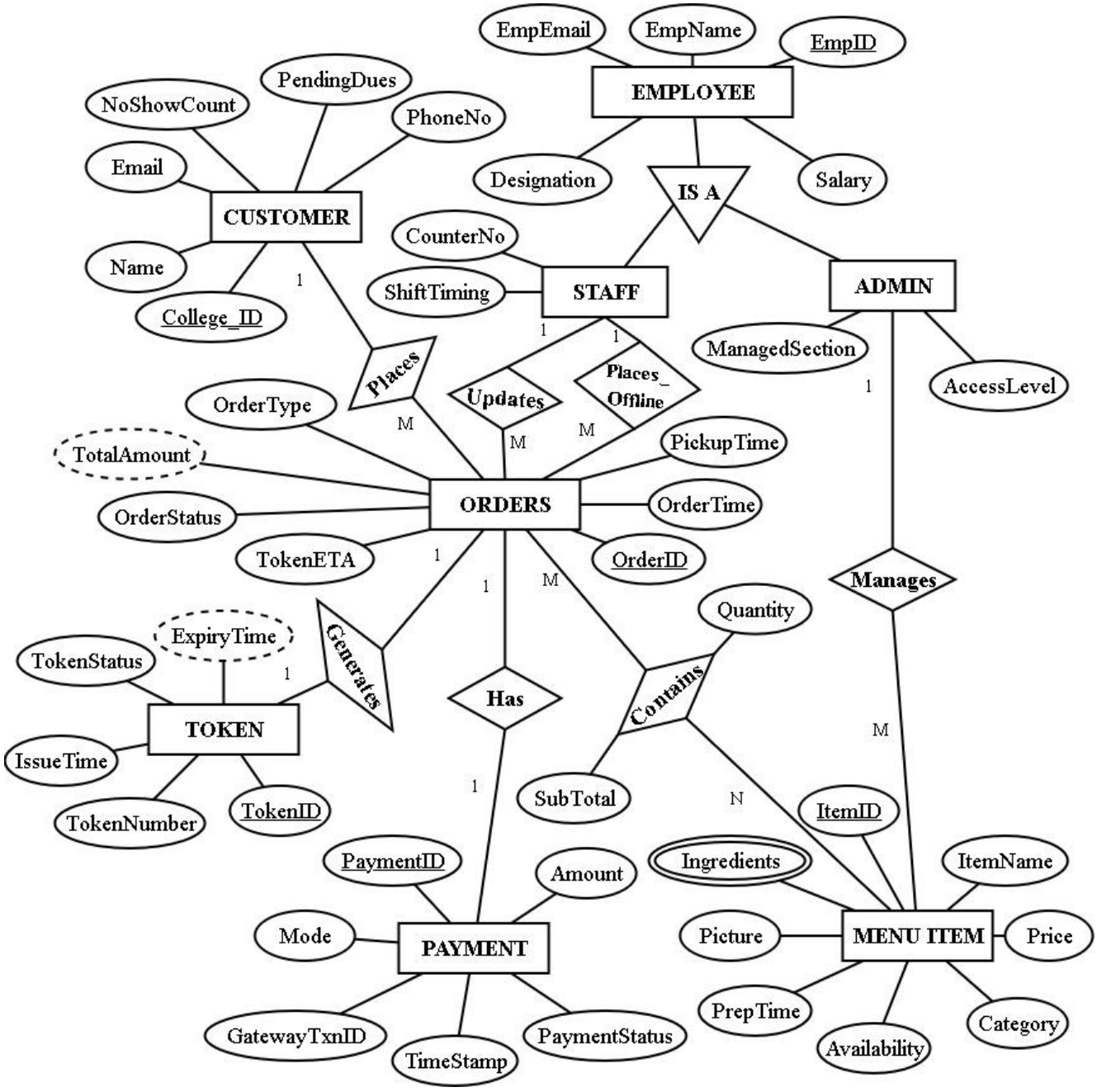


Figure 1: *ER Diagram*

9.2 Use Case Diagram

Customers register, log in, search, order, cancel and track orders. Staff handle queues and offline orders; Admin manages menus, policies, reports and No-Shows. Included and extended use cases cover payments, notifications and token expiry.

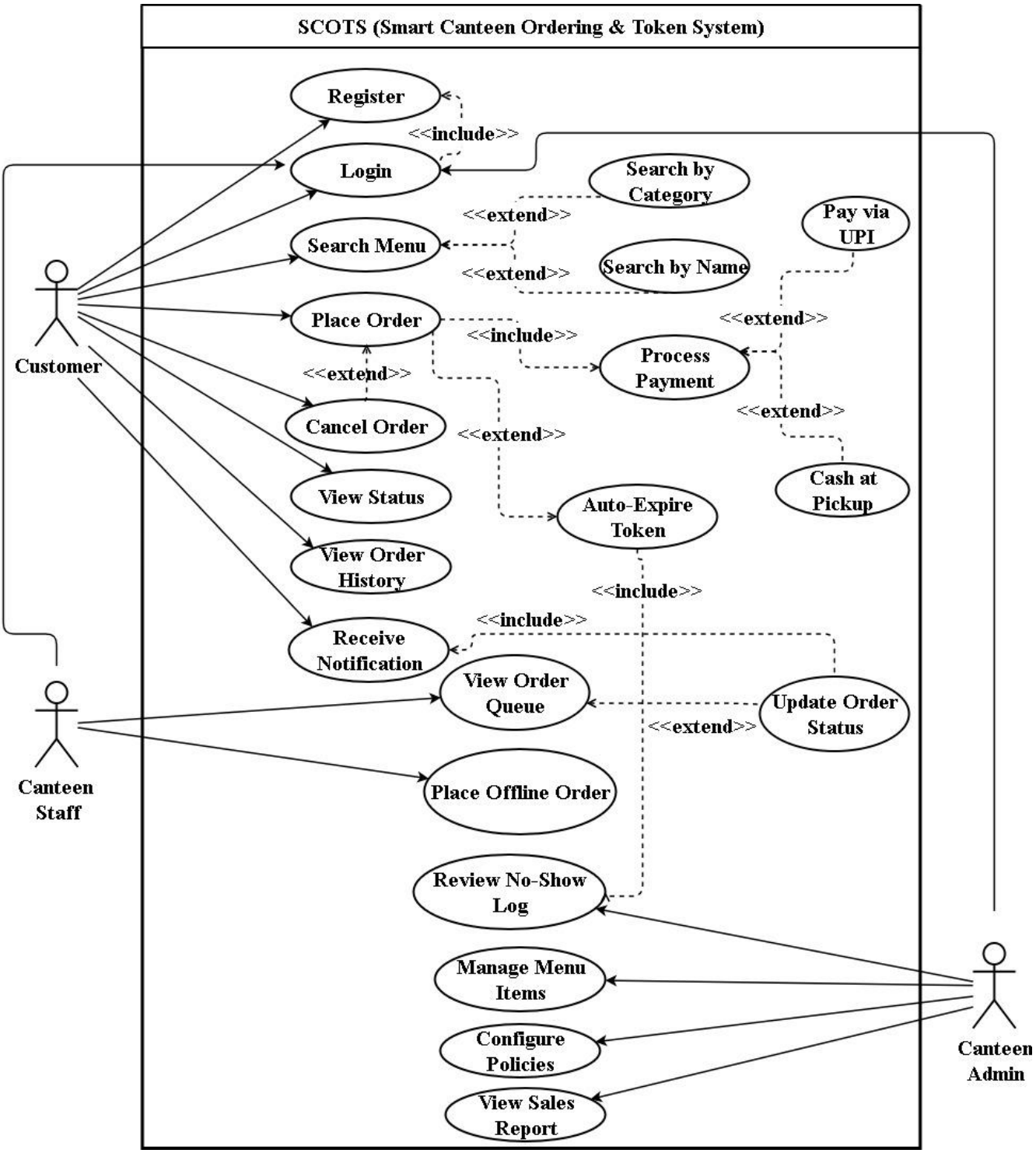


Figure 2: *Use Case Diagram*

9.3 Data Flow Diagrams (DFDs)

9.3.1 Context Diagram (Level 0)

The context diagram shows SCOTS as one process exchanging data with Customer, Canteen Staff, Canteen Admin and Payment Gateway. Flows cover login and order details, token slips, queue alerts, status updates, offline orders, menu updates, reports and payment requests and confirmations.

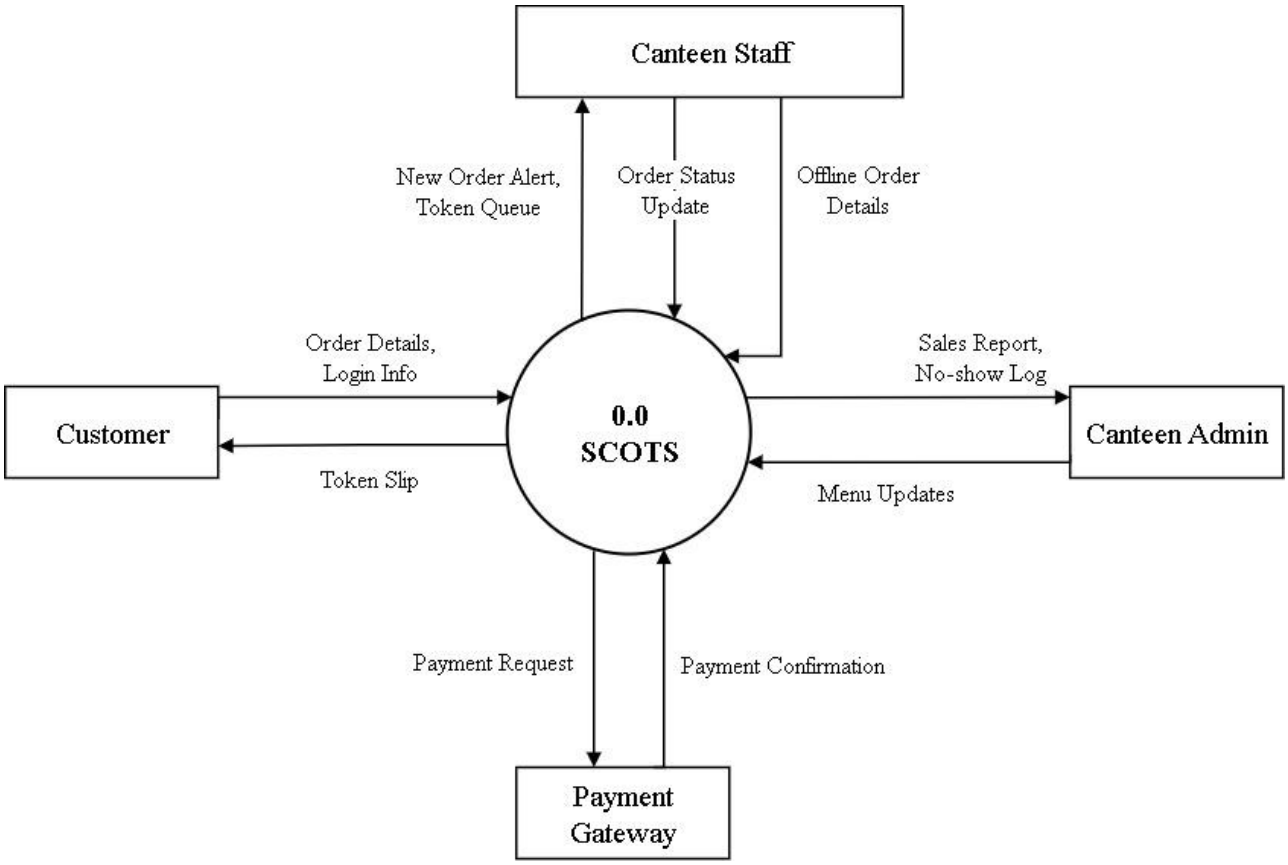


***Figure** 3: Level 0 DFD (Context Diagram)*

9.3.2 Level 1 DFD

The Level 1 DFD connects Manage Menu, Process Order, Handle Payment, Manage Queue and Enforce Policy with Menu, Orders and Users databases. It shows customer and staff order inputs, gateway payment exchanges, token delivery, queue updates, sales reporting, and No-Show logging and restriction checks.

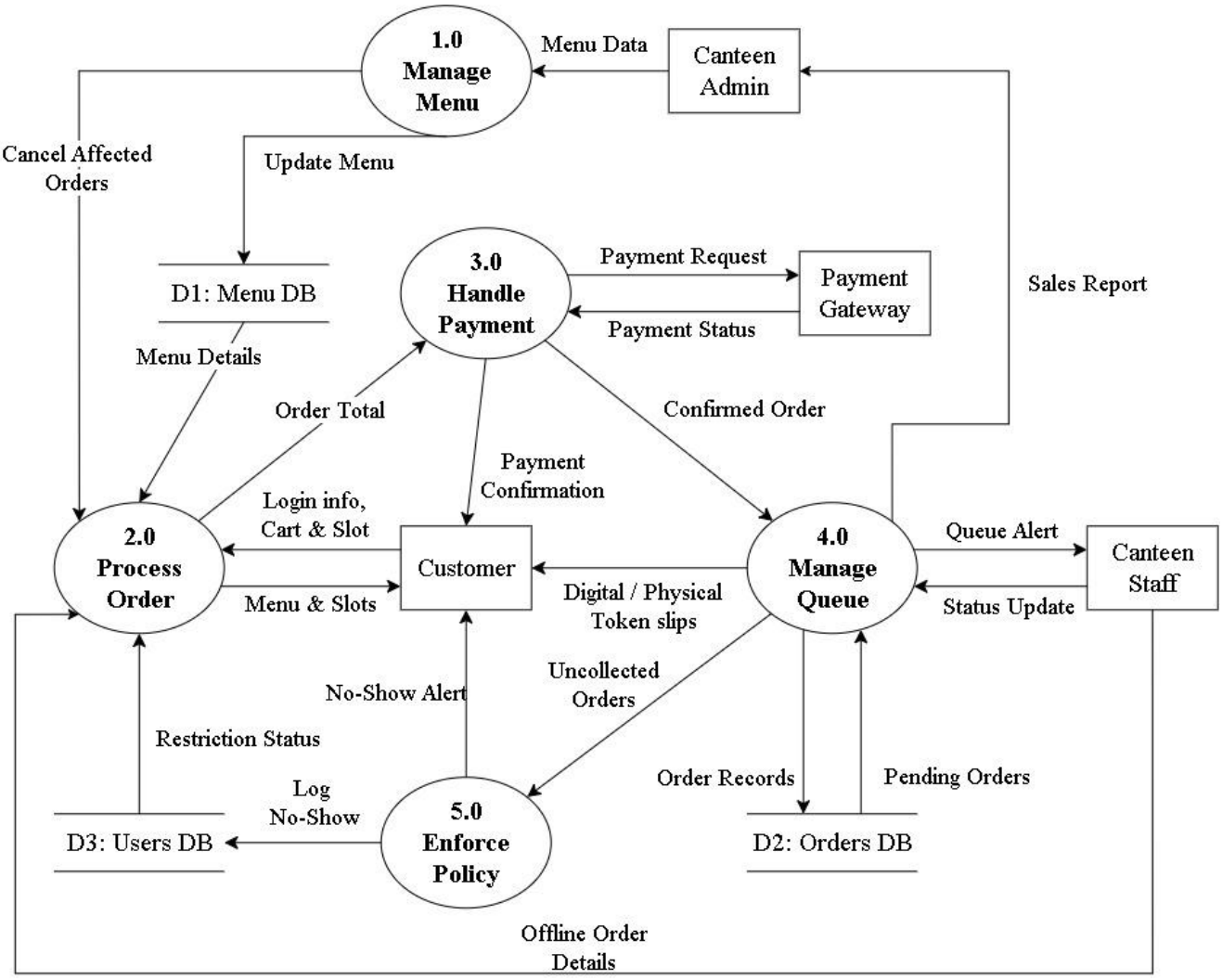


***Figure** 4: Level 1 DFD*

9.3.3 Level 2 DFD (Process 1.0: Manage Menu)

Manage Menu contains Set Price/Prep Time, Add/Edit Item and Mark Sold Out. Canteen Admin supplies prices, preparation times, item details and sold-out flags. Each process updates D1: Menu DB; Mark Sold Out also sends a request to Process Order to cancel affected orders.

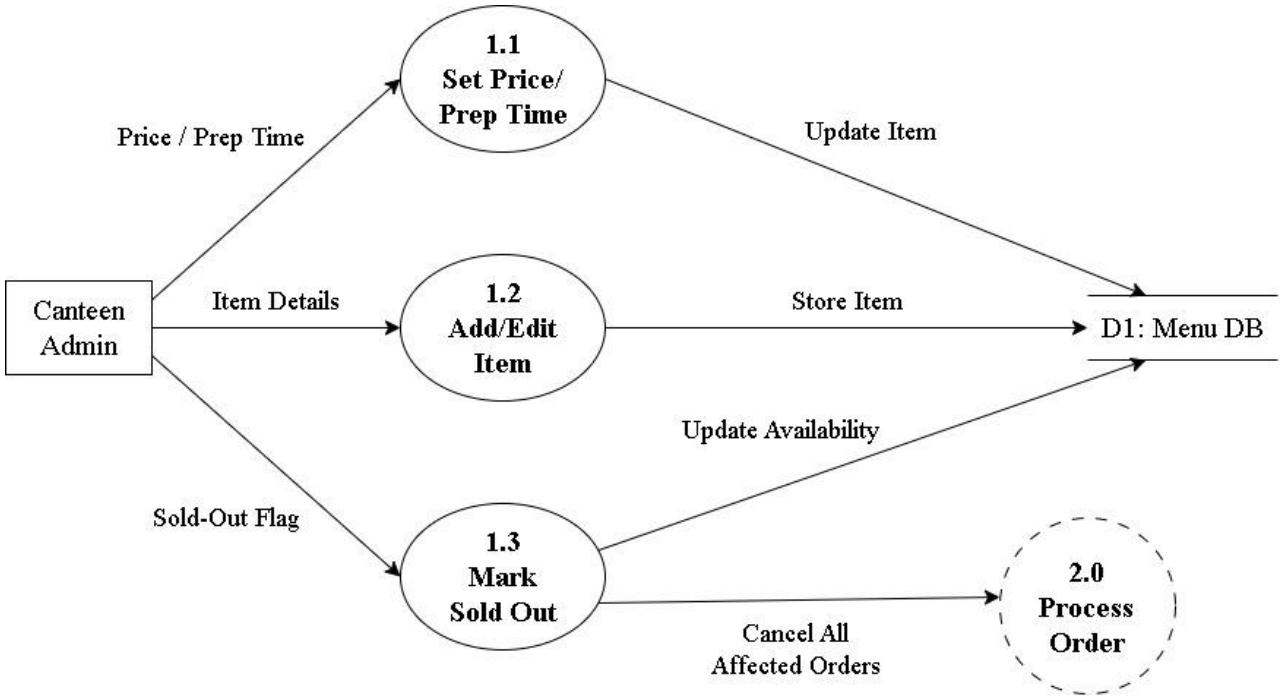


***Figure** 5: Level 2 DFD (Process 1.0: Manage Menu)*

9.3.4 Level 2 DFD (Process 2.0: Process Order)

Process Order contains Build Cart & Checkout ETA, Select Pickup Type, Validate Time Window Capacity, Check Account Restriction and Cancel Order. It uses menu details, customer choices and staff-entered items, reserves capacity in D4, checks D3 and sends the order total to Handle Payment. Cancellation confirms the request, releases capacity and sends a refund/penalty request.

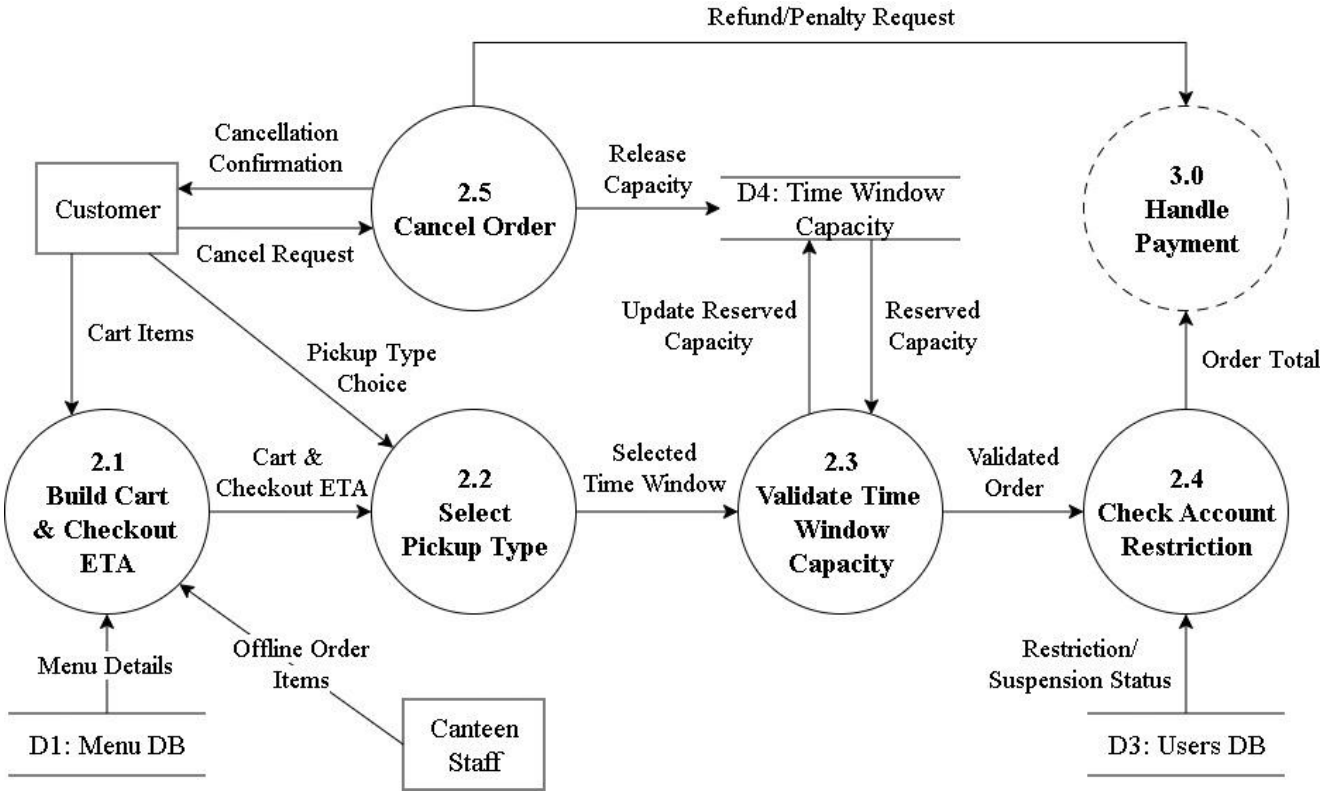


***Figure** 6: Level 2 DFD (Process 2.0: Process Order)*

9.3.5 Level 2 DFD (Process 3.0: Handle Payment)

Handle Payment separates Process Cash-at-Pickup, Process UPI Payment and Process Refund. Cash and UPI flows send confirmed orders to Manage Queue. UPI processing exchanges requests and status with Payment Gateway and confirms payment to Customer. Refund processing exchanges gateway refund data, confirms refunds to Customer and updates Pending Due in D3: Users DB.

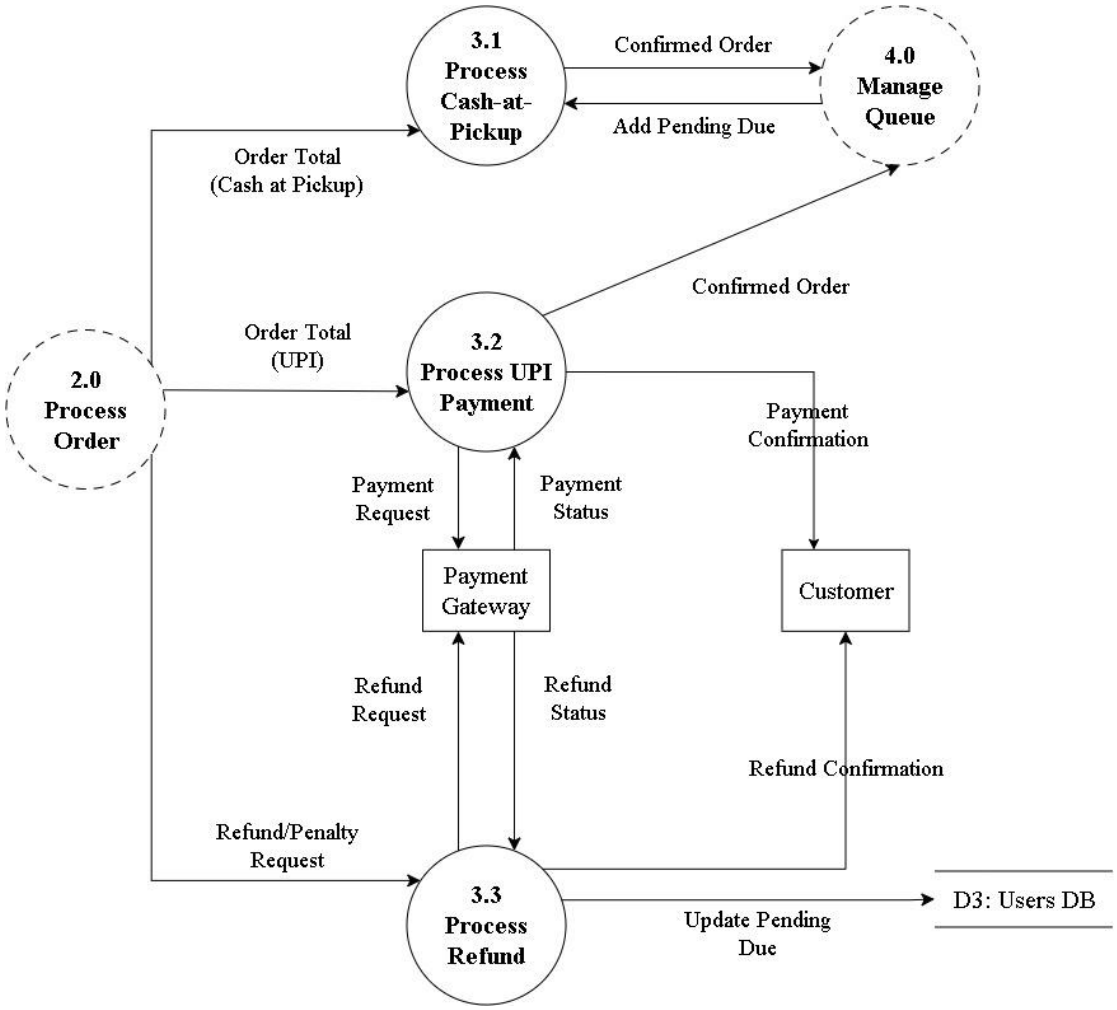


***Figure** 7: Level 2 DFD (Process 3.0: Handle Payment)*

9.3.6 Level 2 DFD (Process 4.0: Manage Queue)

Manage Queue generates tokens and locks ETAs, releases scheduled orders, updates status, expires uncollected tokens, and displays ready and upcoming tokens. These processes use D2: Orders DB. Staff exchange status updates and prompts; Customers receive tokens, ETAs and notifications. Ready and next tokens reach Display Board, while uncollected order information goes to Enforce Policy.

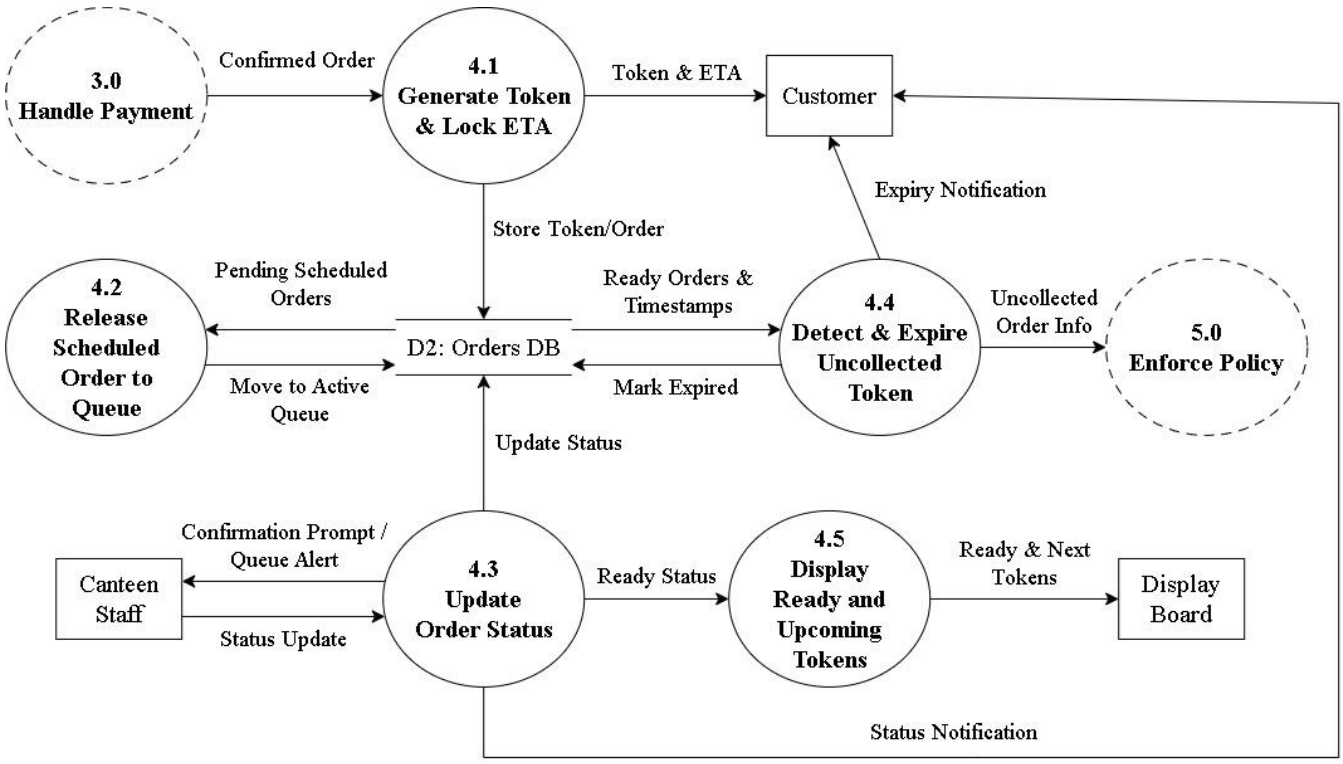


***Figure** 8: Level 2 DFD (Process 4.0: Manage Queue)*

9.3.7 Level 2 DFD (Process 5.0: Enforce Policy)

Enforce Policy logs uncollected orders as No-Shows in D3: Users DB. Evaluate Restriction/Suspension reads 30-day No-Show history, updates account status and triggers Notify Customer. Canteen Admin requests Review & Reinstate, which reads the restricted accounts list and updates D3 to lift restrictions or reinstate accounts.

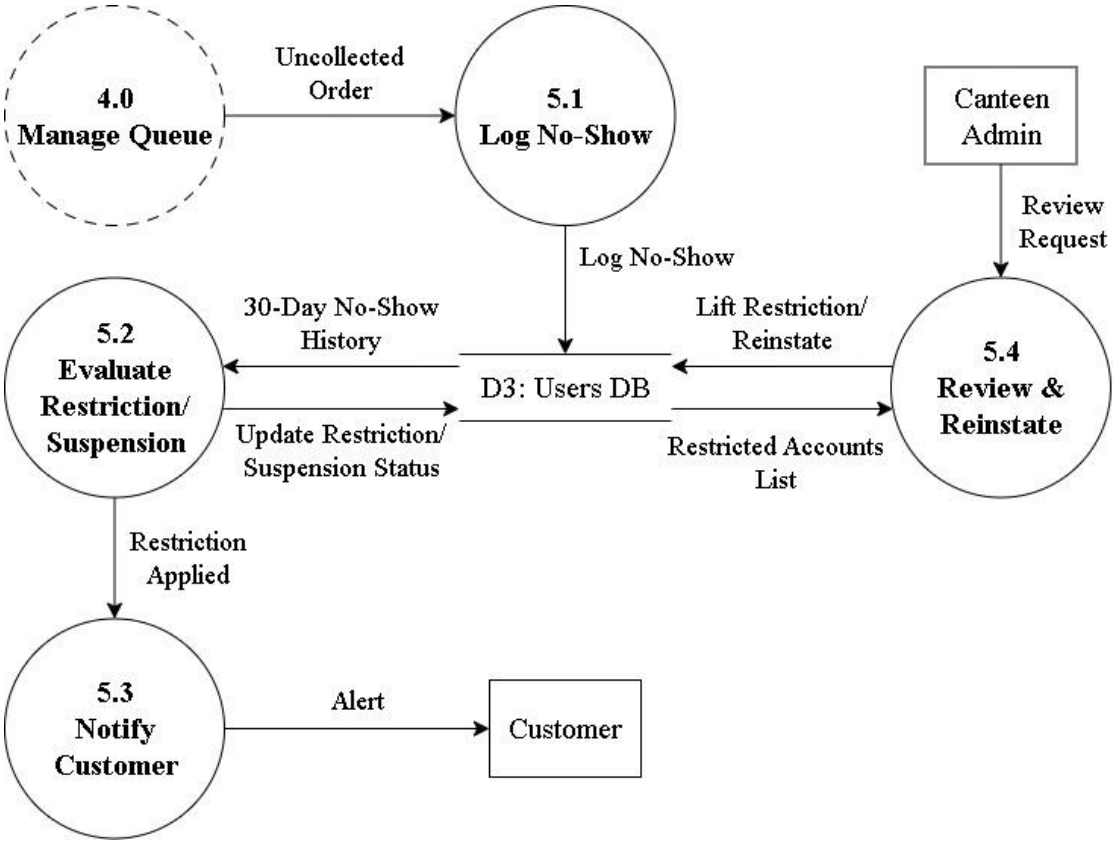


***Figure** 9: Level 2 DFD (Process 5.0: Enforce Policy)*

9.4 Class Diagram

The class diagram shows Admin and Staff inheriting from Canteen Employee. Admin manages Menu Item; Staff updates or places Orders; Customer places Orders. Orders aggregates Menu Item, has a composed Payment and generates a Token. Attributes, methods and multiplicities describe the structure and operations of these classes.

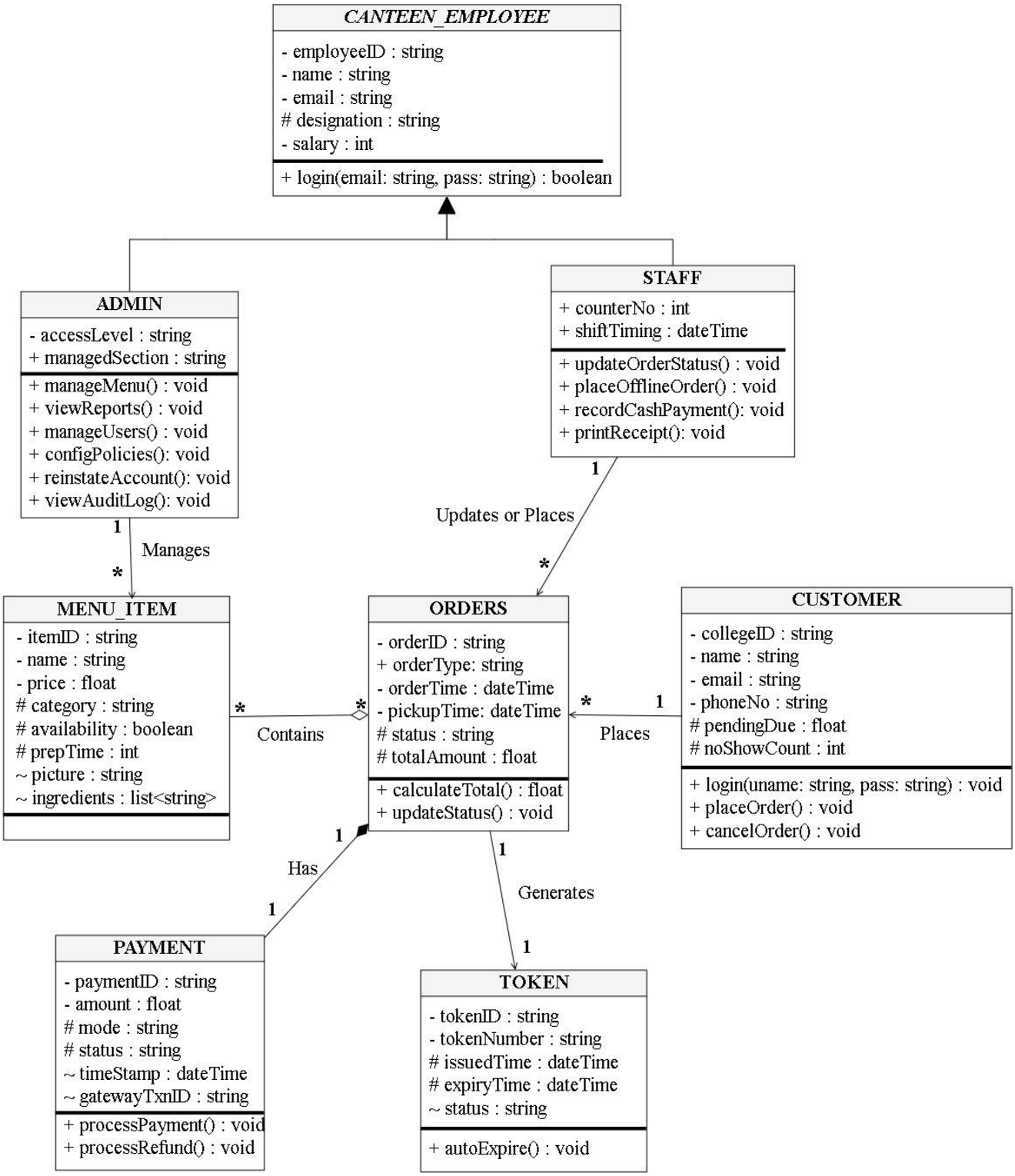


***Figure** 10: Class Diagram*

10. Preliminary Schedule

The table below presents a tentative project timeline over one academic semester (12 weeks), to be refined once detailed design is finalised.

Sr. No.	Activity	Duration	Responsible
1	Requirement Gathering & Analysis	Week 1 – Week 2	Team + Guide
2	System Design (DFD, ER Diagram, UI mockups)	Week 3 – Week 4	Team
3	Database Development	Week 5	Team
4	Backend Module Development	Week 6 – Week 8	Team
5	Frontend/App Development	Week 6 – Week 8	Team
6	Integration, Payment Gateway Testing & Apple Developer Enrollment	Week 9	Team
7	System Testing, iOS Acceptance Testing & Bug Fixing	Week 10 – Week 11	Team
8	Deployment & Demo	Week 12	Team + Guide

11. Future Scope

A few related ideas were considered during requirement analysis but deliberately deferred beyond the current scope, mainly due to the added design and testing complexity:

- Multi-day Advance Booking: Allowing a Customer to pre-order a meal for a future date (e.g., ordering today for pickup tomorrow), which would require calendar-based Time Window management rather than same-day queue management.
- Near-Class/Hostel Delivery: Adding a delivery option beyond counter pickup, which would require delivery-partner assignment and location tracking, outside the current scope (Section 1.2).
- Loyalty/Rewards Program: Offering points or discounts for frequent or on-time-collection Customers, which could also serve as a positive incentive to reduce No-Shows.
- Reuse for Other Campus Services: As noted in Section 4.5 (Reusability), the core ordering/token/queue modules could be adapted for a library book-issue counter or a stationery/print-shop counter with minimal modification.
- AI-Based Demand Forecasting: Historical order data can be used to predict peak hours and popular items on a given day, allowing the canteen staff to prepare inventory and staffing in advance and allowing the system to give more accurate Checkout ETA and Token ETA predictions.

Appendix A: Troubleshooting

This appendix lists common questions and issues Customers, Canteen Staff, and Canteen Admin may encounter while using SCOTS, along with the expected resolution.

Q: My payment was deducted but I didn't get a token.

A: Check the payment status on the Order Status screen. Do not make a second payment for the same attempt while it is pending. The system will reconcile the transaction and either confirm the order once or initiate a full refund if the order cannot be accepted. If the status remains unresolved, contact the counter with the displayed payment reference.

Q: I couldn't collect my order in time; can I get a refund?

A: For an expired prepaid order, the configured percentage of the paid amount (default 100%) is forfeited and the remainder is refunded. For an expired Cash-at-Pickup order, the configured percentage (default 100%) of the order amount is added to the Customer's Pending Due. Contact the Canteen Admin if there was an exceptional circumstance.

Q: The app isn't loading or the Checkout ETA isn't showing.

A: Check the Wi-Fi/internet connection. The app requires an active connection to place orders and to receive real-time order status and token updates (see Section 3.2, Technical Issues).

Q: I've been restricted from using "Cash-at-Pickup" or my account is suspended, how do I get this lifted?

A: A prepayment-only restriction is lifted automatically when the No-Show count falls below two within the configured lookback period (default 30 days). A suspension requires Canteen Admin review and reinstatement. The Admin may also review a restriction under FR23.3. Clear an outstanding Pending Due separately under FR22.1; this does not remove unrelated blocks.

Q: Can I cancel my order after placing it?

A: Yes. An order can be cancelled before it reaches "Ready." The refund or penalty depends on the order status and payment method. Cancelling the configured number of orders (default two) at the "Received" stage within a single operating day temporarily disables ordering for 24 hours (FR20.4).

Q: My order was cancelled because an item sold out. Will I get a refund?

A: Yes. Orders with a sold-out item that are still at the Received stage are cancelled automatically, and any prepaid amount is refunded in full. An unpaid Cash-at-Pickup order is simply voided.

Appendix B: Design Notes

This appendix details the underlying formulas used by the current system to enforce scheduled ordering constraints, keeping the primary Functional Requirements focused on pure outcomes.

B.1 Capacity Model (Refers to FR12.3): N is the configured number of cooks (minimum 1, default 2, maximum 5) and remains constant during an operating day. For a Time Window of length W minutes, physical capacity is **$N \times W$** preparation-work minutes. The Canteen Admin configures Scheduled Order capacity as a share s of physical capacity, where **$0 < s < 1$**; Immediate/Offline capacity is the remainder. **Scheduled Order capacity = $N \times W \times s$ preparation-work minutes**. By default, **$s = 2/3$**. For example, with **$N = 2$** and **$W = 15$ minutes**, physical capacity is 30 preparation-work minutes, Scheduled Order capacity is 20 preparation-work minutes, and Immediate/Offline capacity is 10 preparation-work minutes. The system shall validate that Scheduled Order capacity is greater than zero and strictly less than physical capacity, and shall reject reservations that exceed configured Scheduled Order capacity.

B.2 Availability Validation Formula (Refers to FR12.4): For each Time Window, **Available capacity = Configured Scheduled Order capacity − Confirmed Scheduled Order workload − active temporary reservations**. A new order may be reserved only when its preparation workload is no greater than the available capacity. The check and reservation shall be atomic. Releasing an order into the active queue shall not return its Scheduled Order capacity. A cancelled unreleased order returns capacity under FR20.3; a released order does not.

B.3 Dynamic Queue Integration Calculation (Refers to FR12.5): For each cook, the scheduler stores the time the cook becomes free. A new order is assigned to the cook who becomes free earliest; its predicted start is that time and its predicted end is **start plus preparation duration**. One order stay on one cook, and staff do not assign cooks. The schedule is recalculated whenever an order is placed, becomes Ready or is cancelled. For a Scheduled Order: **Promised Release Time = Time Window start − preparation time** (locked at placement); **Latest Start Time = Promised Release Time + Max Slack**; the order is released into the active queue at its Promised Release Time; **delay = actual kitchen start − Promised Release Time**. An Immediate/Offline Order is placed ahead of a Scheduled Order only if that order can still start by its Latest Start Time; otherwise, it is placed after it and the full simulation is re-run, repeating for any later Scheduled Order affected. A delay beyond Max Slack can occur only through actual kitchen overrun. Before actual start, predicted times are estimates, not delay. Release and expiry timers are persisted and restored after a restart (FR21).

B.4 Capacity Release on Cancellation (Refers to FR20.3): When a Scheduled Order is cancelled before release into the active kitchen queue, subtract its preparation workload from the confirmed reservation for its Time Window and make that capacity available immediately. When an order has started or entered the active kitchen queue, do not return its Time Window

capacity. Remove the cancelled order from the kitchen simulation and recalculate affected ETAs.

B.5 Policy Settings and Defaults (Refers to FR24): The Canteen Admin may configure the following settings; the defaults apply until changed.

Setting	Default value
Time Window length	15 minutes
Kitchen cook count N	2 (integer 1 to 5)
Immediate/Offline preparation-time limit	12 minutes (not above Max Slack)
Scheduled Order capacity share	2/3 (strictly between 0 and 1)
Max Slack	15 minutes (not above Time Window length)
Delay Grace	15 minutes (not above Max Slack)
Minimum Scheduled Order lead time	30 minutes
No-Show forfeiture, prepaid and Cash-at-Pickup orders	100%
Late-cancellation penalty	100%
Offline Order Preparing-stage cancellation and No-Show retention	100%
Auto-expiry window	20 minutes
No-Show lookback period	30 days
Pending Due review period	3 calendar days
Rapid-cancellation threshold	2 Received-stage cancellations per operating day (24-hour block)
Time Window capacity-reservation expiry	3 minutes
ETA reconfirmation tolerance	2 minutes
Required receipt content	As specified in FR11.1

Appendix C: References

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